

Diego Maximiliano Macall

Sustainability | Bioeconomy | Food Systems | Data Science
🌐 Personal Website | 🆔 ORCID | 🐙 GitHub | 📷 Instagram | 🎵 Bluesky

Professional Profile

Multidisciplinary operations, sustainability, and data professional with a background in Agronomic Engineering, Agricultural Economics, and Environmental Science and Technology. Experienced in coordinating international projects, managing complex stakeholder workflows, analysing agri-food and bioeconomy systems, developing analytical tools, and translating data into practical decisions. Strong working knowledge of agriculture, food systems, sustainability, regulatory frameworks, project timelines, budgeting, reporting, and data-driven process improvement. Fluent in English and Spanish, with experience working across academic, government, international, and multicultural environments.

Core Competencies

Operations & Project Coordination

Project planning, timeline management, event operations, workflow coordination, stakeholder follow-up, operational documentation, programme scheduling, and delivery under deadlines.

Logistics, Supply Chains & Food Systems

Agri-food systems, agricultural production, trade flows, food consumption, ethanol trade, bioeconomy systems, circular economy, resource flows, storage and distribution logic, and sustainability constraints across food and agricultural value chains.

Budgeting & Decision Support

Budget preparation, scenario analysis, cost-benefit thinking, investment evaluation, operational reporting, KPI design, dashboard development, and evidence-based recommendations for management decisions.

Data, Statistics & Advanced Quantitative Analysis

Python, R, Excel, statistical analysis, econometrics, survey design, data cleaning, data visualisation, machine learning, time-series analysis, linear programming, optimisation, and applied mathematical modelling.

Microsoft Office & Reporting Tools

Excel for data analysis, tables, charts, and budget tracking; Word for reports, briefs, and formal documents; PowerPoint for executive and technical presentations; Power BI for dashboard-oriented reporting.

Regulations, Policy & Compliance Awareness

Agricultural policy, biotechnology regulation, innovation regulation, municipal food policy, international agricultural trade, regulatory frameworks, access and benefit sharing, and the interpretation of technical and policy documents.

Leadership & Professional Strengths

Attention to detail, fast learning, multidisciplinary thinking, calm performance under pressure, clear written communication, public speaking, multilingual collaboration, stakeholder diplomacy, and the ability to translate technical complexity into practical action.

Professional Experience

Secretary-General / ConfTool Administrator

International Consortium on Applied Bioeconomy Research (ICABR)

February 2018 - Present

- Coordinate operational and administrative workflows for an international bioeconomy research and policy network.
- Helped organize nine international academic conferences, supporting venue preparation, programme scheduling, participant coordination, deadlines, documentation, and travel visa letters.
- Manage the ConfTool platform for submissions, conference administration, participant information, and programme organisation.
- Support communication with senior academics, policy experts, institutional partners, authors, and international stakeholders.
- Coordinate timelines and deliverables for ICABR activities, including annual conferences, communications, publication-related tasks, and strategic initiatives.
- Developed data-driven communication and reporting outputs to translate technical research into accessible decision-oriented summaries.

- Built and maintained analytical tools and structured workflows to support planning, tracking, and organisational decision-making.

Research Assistant

Department of Agricultural and Resource Economics, University of Saskatchewan

September 2017 - November 2021

- Conducted applied research on agricultural biotechnology, food systems, international trade, innovation, and regulatory frameworks.
- Collaborated with research teams to design surveys, collect data, clean datasets, analyse results, and publish peer-reviewed publications.
- Worked with scientists and policy-oriented collaborators in developing countries on agricultural innovation and biotechnology studies.
- Analysed regulatory and market environments affecting agricultural technologies, production systems, and trade.
- Produced written outputs requiring precision, evidence synthesis, technical accuracy, and attention to detail.
- Co-authored peer-reviewed articles, technical reports, and research-based communication outputs.

English-Spanish Translator / Interpreter

Saskatoon Open Door Society

August 2018 - March 2020

- Translated documents between English and Spanish with attention to accuracy, tone, context, and confidentiality.
- Interpreted Spanish-English communication in real time, including during criminal court proceedings.
- Strengthened ability to work under pressure, process complex information quickly, and communicate clearly across languages and institutional settings.

International Agricultural Intern

United States Department of Agriculture, Foreign Agricultural Service

April 2017 - September 2017

- Conducted analysis of ethanol trade between the United States and Brazil, linking agricultural markets, renewable fuels, and international trade dynamics.
- Drafted City Spotlight reports for Sao Paulo, Rio de Janeiro, Belo Horizonte, and Porto Alegre.
- Contributed to the ATO Sao Paulo Literature Review for the EMP-funded Brazil 2040 project.
- Developed experience in policy-oriented reporting, trade analysis, market intelligence, and government-facing communication.

Research Assistant

Ministerio de Educacion de El Salvador

October 2013 - February 2014

- Authored a Spanish-language report on the agronomic assessment of *Bixa orellana* plantations.
- Supported the Salvadoran government's policy interest in scaling up national production of the crop.
- Combined field-oriented agronomic knowledge, policy analysis, and technical report writing.

Selected Applied Projects

ICABR Budget and Planning Tool

- Developed an R Shiny tool to support budget planning, scenario analysis, and decision-making for international conference organisation.
- Applied data structuring, cost tracking, and user-oriented dashboard design to support operational planning.

Spanish Food Consumption Data Dashboard

- Developing a data dashboard to analyse long-term food consumption trends in Spain by region and food group.
- Uses data cleaning, visualisation, and food-system analysis to convert complex datasets into accessible insights.

Climate Change and Nutritional Quality Analysis

- Analysing how climate change may affect the nutritional quality provided by vegetables consumed in Spain.
 - Combines food-system knowledge, environmental science, statistics, optimisation, and policy relevance.
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Education

PhD in Environmental Science and Technology

Universitat Autònoma de Barcelona, Barcelona, Spain

January 2022 - November 2025

- Winner of the prestigious “la Caixa” Doctoral INPhINIT Fellowship.
- Thesis: *From Community Gardens to Rooftops: Practices, Innovation, and Regulations to Enhance Barcelona’s Food System*.
- Advanced knowledge of linear programming, food systems, municipal government, agricultural policy, sustainability, urban agriculture, and regulatory questions affecting food-system innovation.

Master of Science in Agricultural Economics

University of Saskatchewan, Saskatoon, Canada

September 2014 - November 2016

- Thesis: *Ex-Ante Impact Assessment of GM Maize Adoption in El Salvador*.
- Advanced knowledge of agricultural policy, innovation regulation, international agricultural trade, economic impact assessment, and applied quantitative methods.

Agronomic Engineering

Catholic University of El Salvador, Santa Ana, El Salvador

January 2008 - November 2012

- Trained in agricultural production systems, crop management, agronomy, resource use, and agricultural development.
- Winner of the GIZ Essay Contest, supporting participation in the VII Central American Forestry Congress in Managua, Nicaragua.

Technical Skills

Programming & Data Analysis: Python, R, Excel, Power BI, statistical analysis, econometrics, data cleaning, data visualisation, machine learning, time-series analysis.

Dashboards & Tools: R Shiny, D3.js, GitHub, HTML/CSS, dashboard design, executive reporting, structured data workflows.

Geospatial Analysis: QGIS.

Mathematics & Modelling: Linear programming, optimisation, applied statistics, economic modelling, impact assessment, scenario analysis.

Microsoft Office: Excel, Word, PowerPoint, Power BI.

Writing & Communication: Reports, policy briefs, technical writing, academic papers, slide presentations, stakeholder communication, bilingual documentation.

Selected Training, Conferences & Workshops

- From Messy Documents to Structured Data: Practical Text Processing & Analysis for Social Science with Python, February 2026.
- R Package Development in Positron, January 2026.
- Advances in Complex Systems: Addressing the Zero-Emission Goal for Urban Well-Being, Como, July 2024.
- Artificial Intelligence: A Practical Introduction Summer School, University of Warwick, July 2023.
- Machine Learning Summer School, Department of Economics, University of Oxford, September 2022.
- UNICA Ethanol Summit, Sao Paulo, Brazil, July 2017.
- Grow Canada Conference, Calgary, Canada, November 2017.
- GIFS: Emerging Technologies for Global Food Security, Saskatoon, Canada, June 2018.
- ICABR Conferences: Bologna, Buenos Aires, Ravello, and other annual meetings.

Selected Publications

Complete publication list available at: <https://diegomacall.github.io/research.html>

- Macall, D. M., Durany, X. G., Villamayor-Tomas, S., & i Sánchez, D. R. (2026). A food system innovation: Vegetable production in rooftop greenhouses in Barcelona. *Cities*, 171, 106698.
- Macall, D. M., Durany, X. G., & Villamayor-Tomas, S. (2025). Agronomic and organizational aspects of Barcelona’s community gardens. *Frontiers in Sustainable Food Systems*, 9, 1459265.

- Macall, D. M., Madrigal-Pana, J., Smyth, S. J., & Arias, A. G. (2023). Costa Rican consumer perceptions of gene-editing. *Heliyon*, 9(8).
- Macall, D. M., Kerr, W. A., & Smyth, S. J. (2022). Economic surplus implications of Mexico’s decision to phaseout genetically modified maize imports. *GM Crops & Food*, 13(1), 388–401.

Affiliations & Awards

- Secretary-General, International Consortium on Applied Bioeconomy Research.
- Full Member, Sigma Xi, The Scientific Research Honor Society.
- “*la Caixa*” Doctoral INPhINIT Fellowship, 2022-2025.
- GIZ Essay Contest Winner, 2011.

Languages

- **English:** Native speaker and writer.
- **Spanish:** Native speaker and writer.
- **French:** Conversational.

References

Available upon request.

Last updated: June 2026.